CANDIDATE CASE STUDY · CITY BIKE SHARING

Do we need more bikes?

Hourly demand analysis, a hypothesis test, a prediction model and a planning app for operations

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01 · THE PROBLEM

The CEO’s question

RIDERS SAY

“We can’t find a bike when we need one.”

THE CEO SEES

“Bikes sitting unused all over the city.”

“Should we just buy more bikes?” — how I investigated it:

1. Quality checks and features
2. A testable hypothesis
3. A model of bikes needed
4. An app for operations

02 · THE DATA

Two years of hourly rentals

17,379
hourly records

2011–12
Jan 2011 to Dec 2012

17
columns

189.5
average rentals per hour

Group	Columns	Meaning
Time	yr, mnth, hr, season, weekday, holiday, workingday, dteday	When the hour happened
Weather	weathersit, temp, atemp, hum, windspeed	Pre-scaled to 0–1
Demand	casual + registered = cnt	cnt is the target
ID	instant	Row number only

Source: UCI Bike Sharing Dataset (Capital Bikeshare, Washington D.C.)

What the checks found

Check	Finding	Action
Nulls and duplicates	None found	No action needed
Timeline gaps	165 of 17,544 hours missing	Split by time, never shuffle
Humidity = 0	22 rows, one day (sensor fault)	Fill with training mean 0.624
Windspeed = 0	2,180 rows (12.5%)	Kept; barely affects demand
Heavy rain (weather 4)	Only 3 rows	Too rare; app maps it to rain
Target skew	Median 142, max 977	Model learns log(cnt)
Collinearity	temp ≈ atemp (r = 0.99)	Combine into one feature

04 · EXPLORATION

What drives demand

2 peaks

On working days: 8 am (≈477/hr) and 5–6 pm (≈525/hr). Hour of day is the strongest driver.

+63%

Growth from 2011 to 2012: 144 → 235 average rentals per hour.

49 → 330

Rentals per hour below 5 °C vs at 30–35 °C. Temperature is the main weather effect (r = 0.41).

32% vs 13%

Casual riders' share on non-working vs working days: leisure riders vs commuters.

Feature engineering steps

Step	Why
temp_avg = (temp + atemp) / 2	The two are almost identical (r = 0.99)
Drop season	Overlaps with month (r = 0.83)
Drop weekday	Its effect is captured by workingday
Drop instant, dteday	Row ID and raw date; info already in yr, mnth, hr
Drop casual, registered	They add up to cnt: target leakage
hum = 0 → training mean	Sensor fault; train-only mean avoids leakage
New: low_demand_hour	1 for 11 pm–6 am: 34 rides/hr vs 252 otherwise

Final inputs (10): yr · mnth · hr · holiday · workingday · weathersit · hum · windspeed · temp_avg · low_demand_hour → **target: cnt**

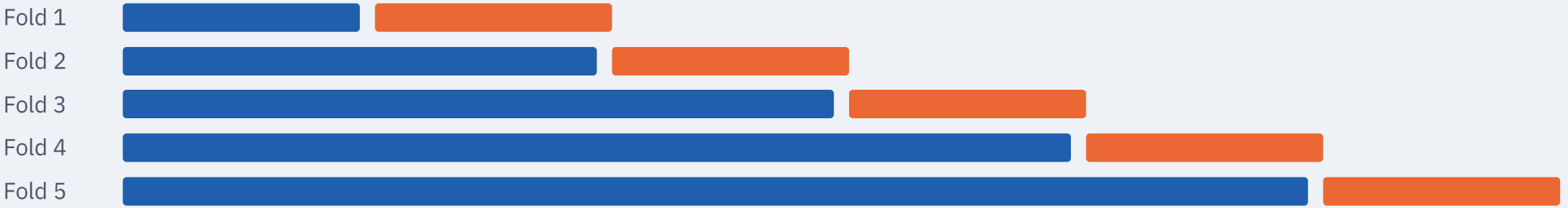
06 · TRAIN / TEST SPLIT

Split by time, never shuffle

Train · 80% · 13,903 hours
Jan 2011 → 7 Aug 2012

Test · 20%
Aug → Dec 2012

Tuning inside training: TimeSeriesSplit, 5 folds



Blue = train on the past · Orange = validate on the next block of time

07 · HYPOTHESIS

Is the shortage about timing?

H1 · WHAT I EXPECT

On working days, commute-peak demand (7–9 am, 5–7 pm) is **more than 2×** midday demand (10 am–4 pm).

H0 · WHAT THE TEST TRIES TO REJECT

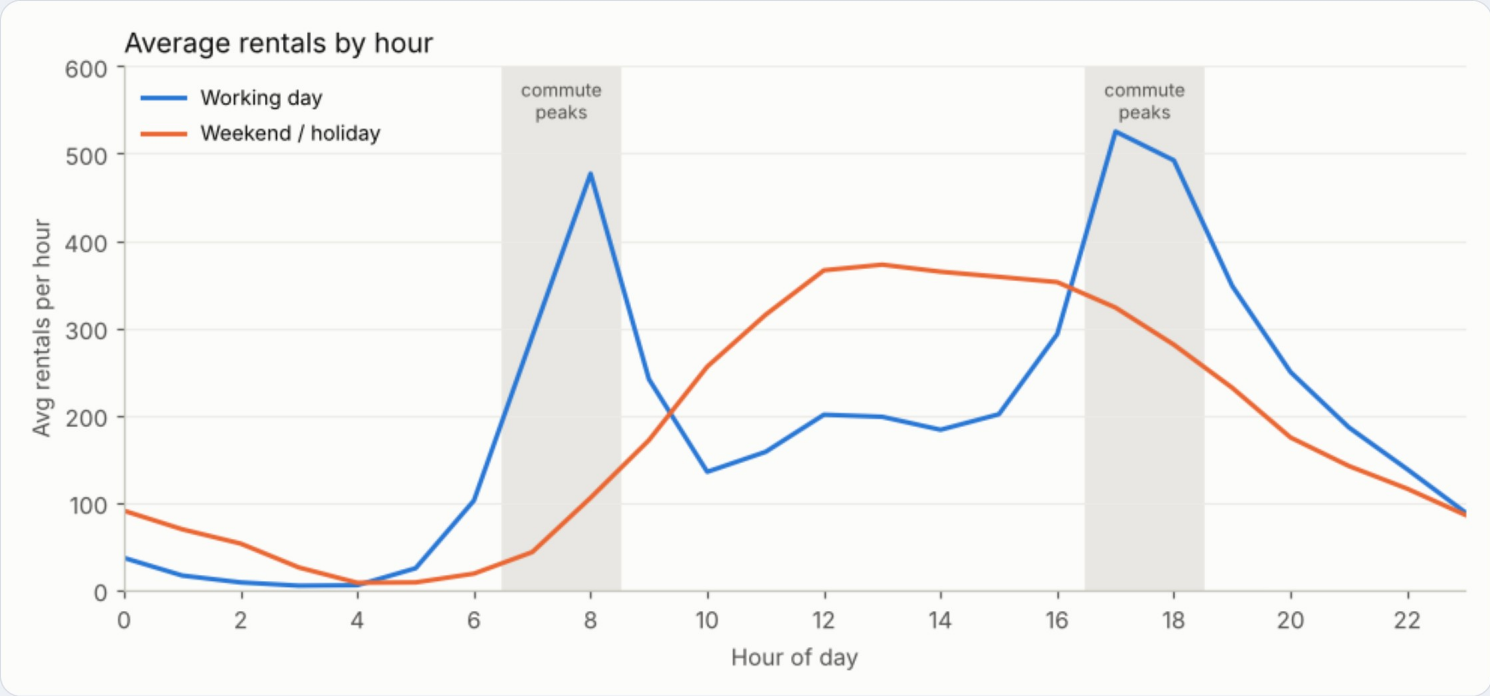
The peak-to-midday ratio is 2 or less.

If H1 holds, a fleet big enough for the peak must sit idle for much of the day.

How I test it

- Unit = one working day (498 days), not one hour
- Ratio per day = peak mean ÷ midday mean
- One-sided Wilcoxon signed-rank test vs 2×
- Bootstrap 95% confidence interval
- Control group: weekends and holidays
- Pass rule set in advance: $p < 0.05$ and CI above 2

Two very different days



WORKING DAYS

Sharp peaks at 8 am and 5–6 pm

WEEKENDS

One smooth midday hill, no rush hour

SO

The peak exists only when people commute

Average rentals per hour, 2011–2012

09 · RESULT

H0 rejected: it is a timing problem

2.62×

Typical peak vs midday demand (95% CI 2.58–2.65)

10^{-52}

p-value: far below 0.05, not luck

0.54×

Same ratio on weekends: the peak vanishes

50%

Of a peak-sized fleet used in an average daytime hour

Robust: holds in both years, all four seasons and every weather type (2.55× to 2.87×).

498 working days and 231 non-working days analysed

Model comparison on unseen months

Model	MAE	RMSE	R^2
Tuned HistGB (chosen)	48.3	74.0	0.887
HistGB (default)	50.4	76.9	0.878
Random Forest	52.2	79.5	0.870
Baseline: hour × day average	101.6	144.9	0.568
Linear Regression	111.4	166.3	0.431

MAE = average error in rentals per hour · Test period: Aug–Dec 2012

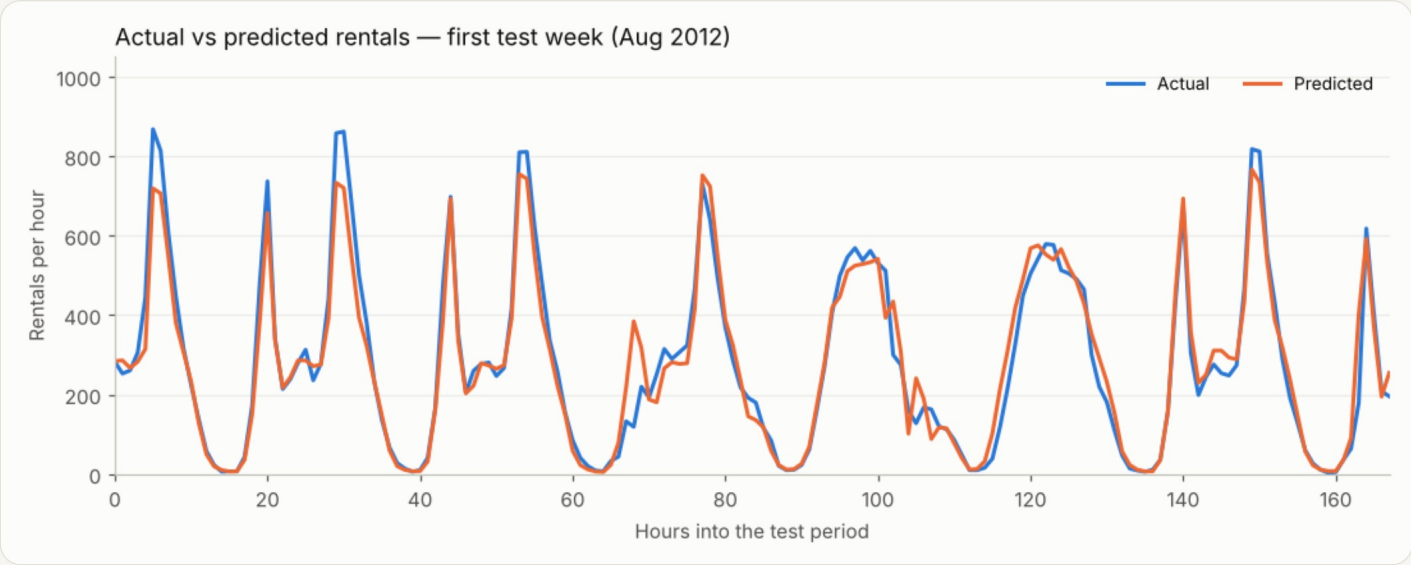
−52%

error vs the simple baseline

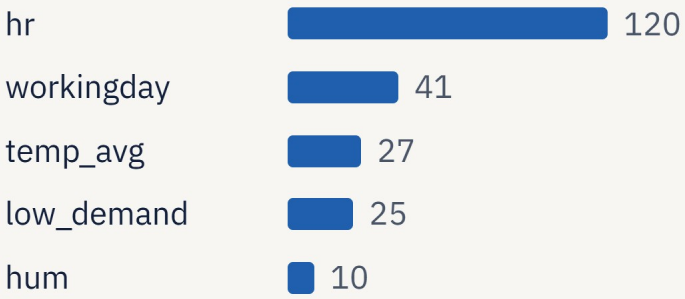
All models learn log(cnt) and are tuned with TimeSeriesSplit.

Linear regression treats the hour as a straight line, so it cannot model two peaks.

It tracks real demand closely



What drives a prediction



Extra error (rentals/hr) when a feature is shuffled; others under 5

⚠ **Leans low at peaks:** under-predicts in 65% of hours (≈114 bikes off at peaks), so plan with a safety buffer.

12 · PROTOTYPE

A planning app for operations

Enter

Date, hour, holiday, and the weather forecast in °C, % and km/h

See

Expected rentals, bikes to have ready (+ buffer), busiest hour, daily chart

Act

Tips: rebalance before commute peaks; use night hours for maintenance

EXAMPLE

Tue 12 Jun 2012, 8 am, clear, 22 °C → **695 rentals** expected, **800 bikes** ready (+15%)

DEPLOYMENT

Streamlit app on GitHub → Streamlit Community Cloud; model + metadata keep preprocessing identical

Don’t buy bikes first. Move them at the right time.

1

Rebalance bikes to commuter stations before 7 am and 4 pm

2

Off-peak pricing to spread demand away from rush hour

3

Plan daily with the model plus a peak buffer; revisit fleet size only if shortages persist

What the data can’t tell us yet

Limitations

- cnt counts completed rentals, so true peak demand is higher
- No station locations or fleet size in the data
- Only 2011–2012; later growth is not captured
- Model leans low at the highest peaks

Next steps

- **Rebalancing pilot:** random stations, 4 weeks, compare empty-station minutes
- **Natural-language assistant** (Groq LLM) for ops questions
- Retrain with recent and station-level data
- Quantile model to plan for 90th-percentile demand